

Ease of oxidation and reduction

Learning outcomes

By the end of this section you should be able to:

- (R3.2.3) Determine the relative ease of oxidation of different metals by analysing redox reactions between metals and aqueous metal ions.
- (R3.2.3) Predict the relative ease of oxidation of metals.
- (R3.2.3) Predict the relative ease of reduction of halogens.
- (R3.2.3) Interpret data regarding metal and metal ion reactions.
- (R3.2.4) Deduce equations for reactions of reactive metals with dilute HCl and H₂SO₄.

In the previous section, you saw a very rusty children's bicycle and you have learned that iron can undergo a redox reaction when in contact with air for a prolonged time. You may also know that nowadays most bicycle frames are made of aluminium. Why is that? Does aluminium react differently to iron when exposed to air?

You learned in the last section what happens when you place zinc metal in a solution of copper ions. But what would happen if you placed copper metal in a solution of zinc ions?

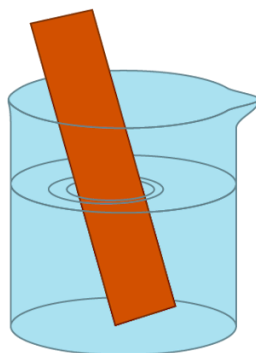


Figure 1. Copper metal in a solution of zinc ions.

No matter how long you leave the copper strip in the zinc ion solution, you will not see anything happen. Why?

Some reactions can proceed in both directions (see [subtopic R2.3 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](#)), allowing the system to reach an equilibrium. This does not happen in these

reactions as they are only evolving in one direction. Why is that?

Relative ease of oxidation of metals

When left in contact with the air, some metals undergo oxidation, which can lead to the formation of metal oxides or other compounds. You learned about the rusting of iron in the previous section and might also have encountered the corrosion of copper, where a greenish-blue colour develops, or the tarnishing of silver, when it loses its shiny appearance.



Credit: Philippe Gerber, Getty Images



Credit: Becart, Getty Images

Figure 2. A corroded copper pipe with its characteristic greenish-blue colour and tarnished silver jewellery, lacking its natural shine.

As seen in **Figure 1**, when a strip of copper metal is placed in a solution of zinc ions there are no visible changes in the solution. However, if you place zinc metal in a solution of copper ions a redox reaction will happen and you will see the blue colour of the solution fade and new copper form around the zinc (**Figure 3**). From these examples, we can deduce that some metals are more reactive than others.

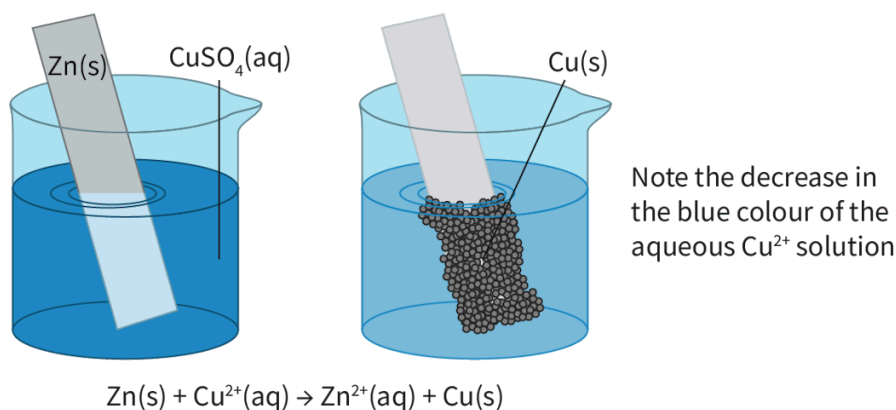
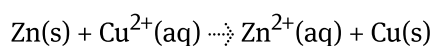


Figure 3. The reaction of zinc metal with copper(II) sulfate solution.

📖 More information for figure 3

The reactions discussed above are known as displacement reactions. In a displacement reaction the metal that is more reactive is able to displace, or replace, the metal ion in the solution.



All metals have a tendency to lose electrons and form positive ions (see [subtopic S2.1 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)](#)). In the equation above, you can see that copper has already lost its electrons (it is a positive ion), but zinc still has them when in its metallic form. From the information provided, you can deduce that zinc has the ability to give its two valence electrons to the copper ion, which will be reduced to copper metal. This means that zinc is oxidised (loses its electrons) more easily than copper, making it a stronger reducing agent. You can also say that zinc is more reactive than copper.

Let's look at a few other reactions:

Table 1. Some displacement reactions.

Met als rea ctin g	Equ atio n	Qualitative observations
Zin c (Zn) wit h lea d (Pb)	$\begin{aligned} &\text{Zn(s)} \\ &+ \\ &\text{Pb}^{2+} \\ &+ \\ &(\text{aq}) \\ &\rightarrow \\ &\text{Zn}^{2+} \\ &(\text{aq}) \\ &+ \\ &\text{Pb(s)} \\ & \\ &\text{Pb(s)} \\ &+ \\ &\text{Zn}^{2+} \\ &(\text{aq}) \\ &\rightarrow \\ &\text{no} \\ &\text{reac} \\ &\text{tion} \end{aligned}$	Zinc Reacts with Lead (II) Nitrate to Produce Lead and Zinc Ni  Video 1. Zinc Reacts with Lead (II) Nitrate to Produce Lead and Zinc Nit 👁 More information for vic
Lea d (Pb) wit h cop per (Cu)	$\begin{aligned} &\text{Pb(s)} \\ &+ \\ &\text{Cu}^{2+} \\ &+ \\ &(\text{aq}) \\ &\rightarrow \\ &\text{Pb}^{2+} \\ &+ \\ &(\text{aq}) \\ &+ \\ &\text{Cu(s)} \\ & \\ &\text{Cu(s)} \\ &+ \\ &\text{Pb}^{2+} \\ &+ \\ &(\text{aq}) \\ &\rightarrow \\ &\text{no} \\ &\text{reac} \\ &\text{tion} \end{aligned}$	Lead and Copper II Chloride  Video 2. Lead and Copper II Chloride 👁 More information for vic

Metals reacting	Equation	Qualitative observations
Copper (Cu) with silver (Ag)	$\text{Cu(s)} + \text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^+(\text{aq}) + \text{Ag(s)}$ $\text{Ag(s)} + \text{Cu}^+(\text{aq}) \rightarrow \text{no reaction}$	Chemistry Experiment - Copper and Silver Nitrate Redox React 

Based on the information provided in **Table 1**, order these metals by increasing reactivity.

Increasing Reactivity →

< < < <

Ease of oxidation →

Zn Ca Cu Ag Pb

✓ Check

Interactive 1. Reactivity Series of Elements.

🔗 More information for interactive 1

Study skills

The relative R of metals observed in metal/metal ion displacement reactions does not need to be learned; appropriate data will be supplied in examination questions.

You learned previously (see [subtopic S3.1 \(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)\)](#) that the periodic table is a useful resource to help us predict some properties of elements. Ease of oxidation is one such property.

Worked example 1

Use the periodic table to determine whether sodium (Na) or potassium (K) shows a more vigorous reaction with water.

Moving down the group, the valence electrons are further away from the nucleus (higher energy level and more shielding), which makes them easier to remove, which leads to increased ease of oxidation. Generally, metals at the bottom of a group are more reactive than metals at the top of that same group.

Worked example 2

Use the periodic table to determine which metals are least reactive. Metals from Groups 1, 2 or 3?

The least reactive metals would be from Group 3 as alkali metals (Group 1) and alkaline earth metals (Group 2) are highly reactive metals that readily oxidise due to their low ionisation energies.

Moving down a group, metals have similar chemical properties and in general the ease of oxidation tends to increase.

Study skills

Lithium is an exception to the general trend of reactivity due to its density and ionisation energy (see [section S3.1.3a \(/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/\)\)](#)).

Worked example 3

Use the periodic table to determine which metal is more reactive. Sodium (Na) or aluminium (Al).

Across a period, the valence electrons are positioned in the same energy level (same shielding), which makes them more attracted to the nucleus due to an increase in effective nuclear charge. These outer electrons are harder to remove, which leads to decreased reactivity and ease of oxidation.

Worked example 4

Use the periodic table to determine which metals are least reactive. Transition metals or metals from Groups 1 and 2?

Transition metals (Groups 3–12) are generally less reactive compared to alkali metals and alkaline earth metals due to their higher ionisation energies and lower reactivity with oxygen.

Metals in the same period may exhibit varying levels of ease of oxidation. In general, metals towards the left side of the periodic table tend to be more easily oxidised than metals towards the right side.

Making connections

The observations may vary depending on the specific metals, halogens, and halide ions involved, as well as the conditions of the reaction, such as concentration, temperature, and presence of catalysts.

Metals mixed with aqueous metal ions

When a more reactive metal is mixed with a solution containing ions of a less reactive metal, a redox reaction occurs, resulting in the more reactive metal displacing the ions of the less reactive metal. This can sometimes be observed as a colour change, such as the change in colour of aqueous copper(II) sulfate when solid zinc is added. This results in a colour change from blue to colourless. Some metals can react with metal ions to produce gas, which can be observed as effervescence or bubbling.

Halogens mixed with aqueous halide ions

If a more reactive halogen is mixed with a solution containing halide ions of a less reactive halogen, a redox reaction occurs, resulting in the more reactive halogen displacing the less reactive halogen. This can result in the change of colour of the solution, the appearance of a precipitate or gas formation. Chlorine solutions are usually a pale green, and bromine solutions are more orange in colour depending on the concentration. Iodine solutions are brown while fluorine solutions are a pale yellow to colourless depending on the concentration.

Relative ease of reduction of halogens

Halogens have a tendency to gain one electron to complete their octet (see [subtopic S2.1 \(/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)\)](#) (**Table 2**).

Table 2. Displacement reactions involving the halogens.

Reaction	Equations	Qualitative observations
Fluorine with chloride ions Chlorine with fluoride ions	$\text{F}_2(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{F}^-(\text{aq}) + \text{Cl}_2(\text{aq})$ $\text{Cl}_2(\text{aq}) + \text{F}^-(\text{aq}) \rightarrow \text{no reaction}$	The colourless solution turns into a pale yellow colour due to the formation of chlorine.
Chlorine with bromide ions Bromine with chloride ions	$\text{Cl}_2(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{aq})$ $\text{Br}_2(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{no reaction}$	Solution changes from pale yellow to orange/brown colour due to the formation of bromine. Chlorine reaction KBr 

Reaction	Equations	Qualitative observations
Bromine with iodide ions	$\text{Br}_2(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Br}^-(\text{aq}) + \text{I}_2(\text{aq})$	Solution changes from orange/brown to da colour due to the formation of iodine.
Iodine with bromide ions	$\text{I}_2(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{no reaction}$	Bromine water reacting with ioc 

Table 2. Displacement reactions involving the halogens.

Use the information provided in **Table 2** to order these metals by increasing reactivity.

The diagram consists of two horizontal arrows pointing to the right. The top arrow is labeled "Increasing Reactivity". Below it, three less-than signs (<) are spaced out. The bottom arrow is labeled "Ease of reduction". Below this arrow, four boxes containing the chemical symbols Cl, I, F, and Br are arranged from left to right. At the bottom left of the diagram area is a "Check" button with a checkmark icon.

Interactive 2. Reactivity Series of Halogens.

For the halogens, moving down the group, there is an increase in atomic radius due to valence electrons being placed further away from the nucleus. This reduces their pull to the nucleus, which also decreases the electrostatic force between the nucleus and the electrons. Hence the ease of reduction is very easy to follow in the periodic table; it decreases moving down the group.

Making connections

Why does metal reactivity increase, and non-metal reactivity decrease, moving down the main groups of the periodic table? (see [subtopic S3.1 \(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](#)).

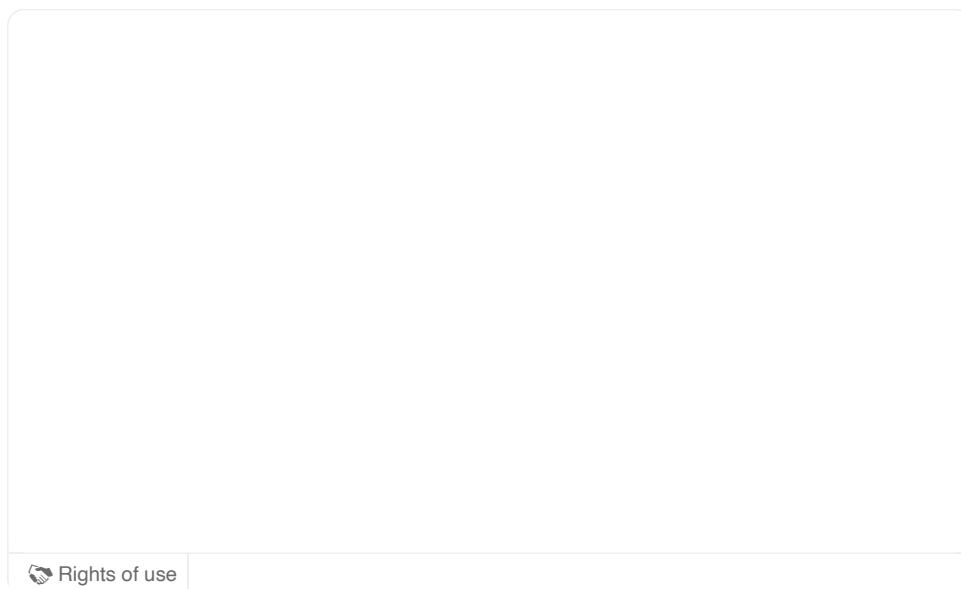
These two trends were discussed in the examples above.

Moving down a group, the valence electrons are placed at higher energy levels, which means there is more shielding between the nucleus and the outer electrons. This decreases the effective nuclear charge, which means the nucleus has less strength to attract the outer electrons. Because of this lower electrostatic force, less energy is required to remove the outer electrons from the elements at the bottom of the group.

For the metals, this means they are more easily oxidised — they can lose their outer electrons more easily. For the non-metals, this means they have less pull on the electrons that they receive to complete their octet — so they are less easily reduced.

Reaction of dilute acids with metals

Now let's look at a reaction between a metal, like magnesium with hydrochloric acid. What do you observe? What do you think is happening?

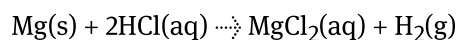


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Video 1. Magnesium Ribbon Placed in Hydrochloric Acid.

 More information for video 1

The equation for the reaction is:



The bubbles that you see in the clip above are the hydrogen gas produced by this displacement reaction. When acids react with certain metals, such as zinc, aluminium and iron, hydrogen gas is produced. In this reaction, the magnesium is oxidised (loses electrons) to form Mg^{2+} ions, and the hydrogen ions (H^+) in the acid are reduced (gain electrons) to form hydrogen gas (H_2). You can test the gas produced by using a lit splint or a flame; if hydrogen gas is present, it will combust and produce the characteristic pop sound.

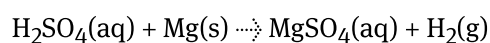
Test for hydrogen gas



Video 2. The test for hydrogen gas. Magnesium can also react with sulfuric acid.

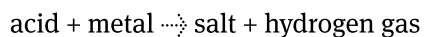
Worked example 5

State the balanced equation for the reaction between magnesium and sulfuric acid.



The hydrogen ion from the acid is displaced by the magnesium which loses two electrons to become Mg^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

The general chemical equation for the reaction between hydrochloric or sulfuric acid and a reactive metal is:



Some common examples of metals that react with acids are:

- Zinc (Zn)
- Iron (Fe)
- Magnesium (Mg)
- Aluminium (Al)
- Tin (Sn)
- Lead (Pb)
- Nickel (Ni)
- Copper (Cu)

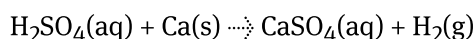
Worked example 6

State the balanced equation for the reaction between hydrochloric acid and zinc.



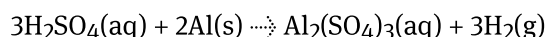
The hydrogen ion from the acid is displaced by the zinc which loses two electrons to become Zn^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

State the balanced equation for the reaction between sulfuric acid and calcium.



The hydrogen ion from the acid is displaced by the calcium which loses two electrons to become Ca^{2+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

State the balanced equation for the reaction between sulfuric acid and aluminium.



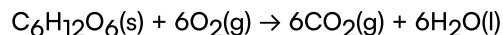
The hydrogen ion from the acid is displaced by the aluminium which loses three electrons to become Al^{3+} and each hydrogen ion gains one electron to form hydrogen gas (H_2).

Aspect: Global impact of science

In the search for renewable, non-pollutant, and sustainable sources of energy, fuel cells were created. You may recall that in fuel cells hydrogen is used as a power source (see [section R1.3.5 \(/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/\)](#)). Hydrogen needs to be produced industrially when acids react with metals, such as magnesium, zinc or iron. The use of these redox reactions has allowed the mass production of hydrogen gas, but not without challenges and risks associated with its production and use (see [section R3.2.6—7 \(/study/app/preview-p/sid-426-cid-753301/book/voltaic-cells-and-rechargeable-batteries-id-46490/\)](#)).

Redox reactions are also essential in ecosystems as they are involved in the cycling of various nutrients. For example, dead organic matter in the soil (or water) will be decomposed by microorganisms through a reaction called cellular

respiration:



This leads to the oxidation of the organic compounds that made up the organism and returns simpler inorganic and/or organic compounds to the ecosystem to be reused.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

You have studied various displacement reactions in this section. Use this [simulation](https://teachchemistry.org/classroom-resources/metals-in-aqueous-solutions-simulation)  (<https://teachchemistry.org/classroom-resources/metals-in-aqueous-solutions-simulation>) to review and observe what happens when you place some metals in:

- a metal ion solution (activities one, two, three)
- hydrochloric acid (activity four).

Create a table where you can write down your results and finish by ordering the metals by increased reactivity.